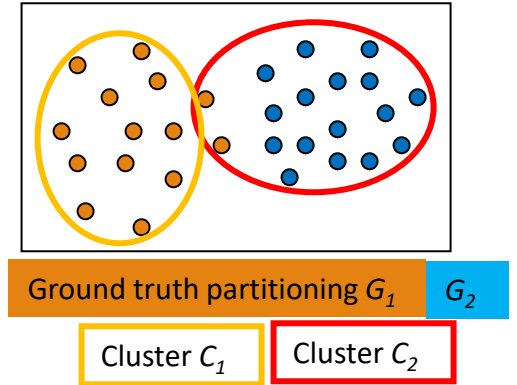


Commonly Used Extrinsic Methods

❑ Matching-based methods

- ❑ Examine how well the clustering results match the ground truth in partitioning the objects in the data set



❑ Information theory-based methods

- ❑ Compare the distribution of the clustering results and that of the ground truth
- ❑ Information theory (e.g., entropy) used to quantify the comparison
- ❑ Ex. Conditional entropy, normalized mutual information (NMI)

❑ Pairwise comparison-based methods

- ❑ Treat each group in the ground truth as a class, and then check the pairwise consistency of the objects in the clustering results
- ❑ Ex. Four possibilities: TP, FN, FP, TN; Jaccard coefficient

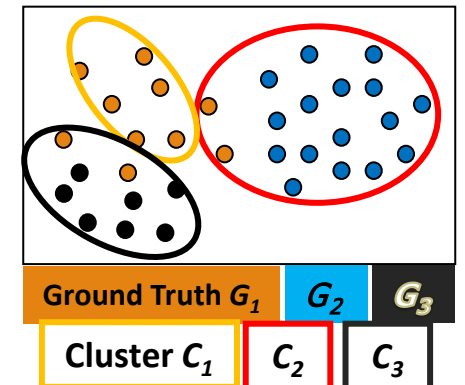
Matching-Based Methods

- ❑ The matching based methods compare clusters in the clustering results and the groups in the ground truth
- ❑ Suppose a clustering method partitions $D = \{o_1, \dots, o_n\}$ into m clusters $C = \{C_1, \dots, C_m\}$. The ground truth G partitions D into l groups $G = \{G_1, \dots, G_l\}$
- ❑ Purity: the extent that cluster C_i contains points only from one (ground truth) partition

❑ Purity for cluster C_i : $\frac{|C_i \cap G_j|}{|C_i|}$, where C_i matching G_j maximizes $|C_i \cap G_j|$

❑ Total purity of clustering C :

$$purity = \sum_{i=1}^m \frac{|C_i|}{n} \max_{j=1}^l \left\{ \frac{|C_i \cap G_j|}{|C_i|} \right\} = \frac{1}{n} \sum_{i=1}^m \max_{j=1}^l \{|C_i \cap G_j|\}$$



Matching-Based Methods: Example

- Consider 11 objects

Object	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>	<i>i</i>	<i>j</i>	<i>k</i>
Ground truth $\mathcal{G}$	G_1	G_1	G_1	G_1	G_1	G_1	G_2	G_2	G_2	G_2	G_3
Clustering $\mathcal{C}_1$	C_1	C_1	C_1	C_1	C_2	C_2	C_3	C_3	C_3	C_3	C_4
Clustering $\mathcal{C}_2$	C_1	C_1	C_2	C_2	C_2	C_3	C_1	C_2	C_2	C_1	C_3

Purity for clustering $\mathcal{C}_1 = 1/11 (4 + 2 + 4 + 1) = 11/11 = 1$;
Purity for clustering $\mathcal{C}_2 = 1/11 (2 + 3 + 1) = 6/11$

- Other methods:

- maximum matching; F-measure

